Computer Vision Techniques for Hand Gesture Recognition: Survey

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Abstract. Hand gesture recognition has recently emerged as a critical component of the human-computer interaction (HCI) concept, allowing computers to capture and interpret hand gestures. In addition to their use in many medical applications, communication between the hearing impaired, device automation, and robot control, hand gestures are of particular importance as a form of nonverbal communication. So far, hand gesture recognition has taken two approaches and relied on a variety of technologies; the first on sensor technology and the second on computer vision. Given the importance of hand gesture recognition applications and technology development today, the importance of the research lies in shedding light on the latest techniques used in the recognition and interpretation of hand gestures. A survey on the techniques used from 2017–2022 has been presented in this research, with a focus on the computer vision approach. The survey was carried out as follows: the first part dealt with research based on artificial intelligence techniques for hand gesture recognition, and the second part focused on research that used artificial neural networks and deep learning for hand gesture recognition.

Keywords: AI Techniques · Deep learning · Human-computer interaction HCI · Sign language · Hand gesture recognition

1 Introduction

According to the notion of human-computer interaction (HCI) technology, humans and machines may interact naturally. Conventional human-machine contact is mostly done through devices like mice, keyboards, and displays. It goes without saying that these devices generally need to be connected to a computer. In some cases, such as virtual reality (VR), remote control (RC), and augmented reality (AR), the procedure of linking these devices is insufficient. As a result, it is critical to do research on how to create an HCI environment that is in tune with human communication behaviors [1, 2].

Gesture recognition is an important and major research topic in the field of assistive technology. The hand gesture is a convenient way to transfer the information as well as an alternative to devices such as the mouse and keyboard. Also, it helps older adults who

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cannot walk or talk communicate with their caregivers when they need help through hand gestures [2].

The use of hand gestures varies from one application to another, depending on the user's cultural background, application domain, and environment. Likewise, with expressive gestures [3]. One of the new paradigms today is the so-called natural user interface (NUI). The basic idea is that the user does not have to rely on additional hardware as a means of input or output and can operate the computer system in the same way that real objects would be input [3, 4].

Recognizing hand gestures has gained increasing importance in recent times for two main reasons: first, the growth of the deaf and hard of hearing population around the world, with a small percentage; and second, the development of applications based on vision and touchless control on devices such as video games, smart TV control, and virtual reality applications [4].

Hand gestures provide a non-physical link to the computer for user comfort and safety, as well as the ability to handle complicated and virtual settings in a much easier manner than the traditional ways in a significant variety of HCI applications [5]. Hand gesture applications, on the other hand, need the ability to correctly apply and comprehend various gestures [6].

1.1 Hand Gesture Methods

Gesture recognition aims to establish a system that can recognize human gestures for data transfer or command and control. It detects human hand motion and translates it into commands [6]. To collect data or information for the identification of hand gestures. This may be accomplished primarily using two techniques [7].

Sensor-Based Approaches. Collecting data consisting of finger and hand position, movement, and trajectory requires the use of sensors or devices physically attached to the user's arm or hand. The main sensor-based methods are:

1. The glove-based approach measures hand and finger position, acceleration, degrees of freedom, and flexion.
2. Electromyography (EMG) measures the electrical impulses of human muscles and decodes biological signals to detect the movement of fingers.
3. Other uses include mechanical, ultrasonic, electromagnetic, and other tactile techniques.

Vision-Based Approaches. With these approaches [8], human movements are captured by one or more cameras [9], and vision-based devices can handle many features to interpret gestures (i.e., color, texture, and shape of the hand). The sensor does not have this characteristic. Although these approaches are simple, they can pose many challenges.

Systems that require a variety of lighting, complex backgrounds, the presence of objects of hand-like skin color (clutter), and some criteria such as detection time, speed, durability, and computational efficiency [8, 9].

The term "vision-based" is most commonly used to capture images and recordings of bare hands without gloves or markers. The sensor-based approach reduces the need for preprocessing, which is the basis of traditional vision-based motion detection systems

[7, 10]. The vision-based approach also doesn't need any extra tools besides cameras, but it does need a lot of data to make a framework that can be used for scenarios that aren't visible (see Fig. 1).

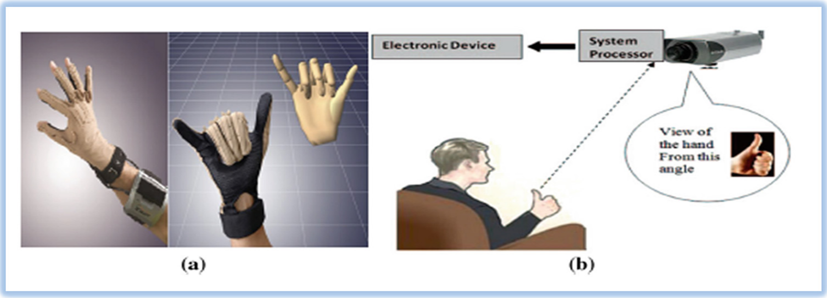

Fig. 1. Hand Gestures. (a) Glove-based attached sensor (b) computer vision–based camera [11].

There are many categories of hand gestures, conversational-gestures, manipulated (metaphoric)-gestures, fingerspelling-gestures, and communication/controlling-gestures [7].

It is very important to focus on the recent techniques that deal with the concept of recognizing gestures under the computer vision approach and to compare the recent work of researchers by highlighting the stages pursued by researchers and surveying their papers for the past five years. All phases of artificial intelligence research into hand gesture recognition were analyzed in this work, along with a summary of deep learning research into hand gesture recognition processes.

1.2 Applications of Gesture Recognition System

Gesture recognition as new interactive technology makes human-machine interactions more natural, convenient, and efficient. The gesture recognition applications are as follows:

1. Talking to the computer: instead of clicking the mouse can add citations or move images by simply flicking the hand [11].
2. Medical surgery: Gestures can be used to control the visual display and assist users with disabilities as part of rehabilitation therapy [2].
3. Gesture-based game control [13].
4. Hand gestures to control home appliances such as MP3 players and TVs, etc. [4].
5. Sign language: Sign language applications are very important in order to bridge the communication gap between deaf people and people who do not suffer from speech and hearing problems [3]. There are many sign languages, most notably American, British, Hindi, Arabic, etc.
6. The driver can be viewed from the eye to choose a radio station, and you can change the station without looking away from the road by simply flicking the hand [6].

7. When the term "immersion" is used, it is frequently associated with augmented reality [12]. Immersive media can create an environment that can be stopped and interacted with. "Immersion" is a term that has attracted much attention in technology (see Fig. 2).

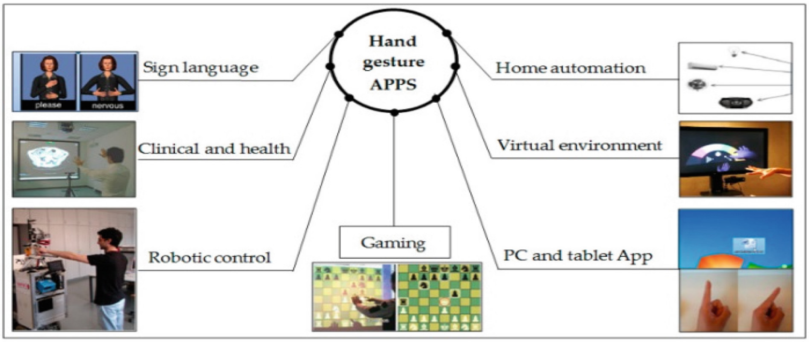

Fig. 2. Application of Hand Gesture [13].

2 Research Aims and Approach

The goal of this paper is to review current issues and advances in vision-based hand gesture recognition, as well as to identify potential future directions. The main research question is: "What are the most recent technological advances and advancements in vision-based hand gesture recognition systems?" To answer this question, all articles for hand gesture recognition systems from 2017 to 2022 have been gathered and identified the methods used. The research was a serious look at each stage of hand gesture recognition. It focused on hand gesture recognition systems based on segmentation, feature extraction, and classification.

The second section will have presented the previous research works that deal with survey research within a specific approach, as well as present a set of data sets and their specifications. The third section presents the methodology of the research sections, where a comprehensive survey will be addressed for each stage of hand gesture recognition based on artificial intelligence and deep learning (see Table 1). The last section is the conclusion view and the references used.

2.1 Related Works

Several reviews summarize a series of studies on gesture recognition. Mention a few pertinent research papers.

[14] research focuses on how to extract structural information and spatiotemporal features in a sequence of video frames by checking three groups based on the type of neural

network used in gesture recognition. Tow-dimensional convolutional networks, three-dimensional convolutional neural networks, and long-term memory (LSTM) networks. Researchers discussed the advantages and limitations of current technology.

[15] focuses on the sub-steps present in each major step associated with system design, namely: data acquisition, segmentation, tracking, feature extraction, and gesture recognition. Therefore, segmentation, tracking, feature extraction, and identification techniques were studied and analyzed. This work reviewed the comparative studies of various technologies for recognizing hand gestures.

[16] is concerned with the provision of sensory glove-based sign language recognition (SLR) systems. The trends and gaps available in this research approach have been explored to provide insights into technological environments. The review looked at the most recent research in four main areas: development, framework, recognizing other hand gestures, and reviews and surveys.

[17] focused on the latest developments in multimedia gesture recognition and presented a JMLR special topic on gesture recognition between 2011-and 2015. It also looks at the latest work on gesture recognition based on a proposed taxonomy for gesture recognition and talks about the problems researchers have had, how they solved them, and where they want to go with their research in the future.

[18] outlines the vision-based hand gesture recognition algorithms reported over the last 16 years. Methods using RGB and RGBD cameras have been validated by quantitative and qualitative comparisons of the algorithms. The algorithm uses a set of 13 metrics selected from the various attributes of the algorithm, an empirical methodology used to evaluate the algorithm, and algorithm detection accuracy to predict success in real-world applications. It will be compared quantitatively. The paper also reviewed the 26 publicly available hand gesture databases and provided web links for download.

[19] Studies centered on the applications of profound learning to RGB-D-based motion acknowledgment highlighted the strategies of encoding spatial-temporal-structural data characteristics in the video. Unlike the above literature, which provides a broad and general report on gesture recognition, this survey covers most of the new hand gesture recognition technologies based on AI and deep learning from 2017 to 2022.

2.2 A Review of Gesture Recognition Dataset

Gesture recognition is a significant focus of HCI research. The gesture can be defined as dynamic or static depending on whether it occurs during a specific time or at a specific instant. In general, dynamic gestures are more diverse and more expressive. Five datasets for gestures that have been largely adopted from publicly available datasets will be mentioned.

20 BN Jester Dataset. Dataset 20 BN-JESTER 1 is a big set of videos labeled using a laptop camera or webcam, and these videos show people making preset hand gestures (see Fig. 3). This dataset can be obtained by visiting this link: https://www.kaggle.com/datasets/toxicmender/20bn-jester/version/1.

Dvs-Gesture Dataset. This dataset was used to create the real-time gesture recognition system described in the paper [A Low-Power, Fully Event-Based Gesture Recognition System]. A DVS128 was used to record the data. The dataset contains 29 people and

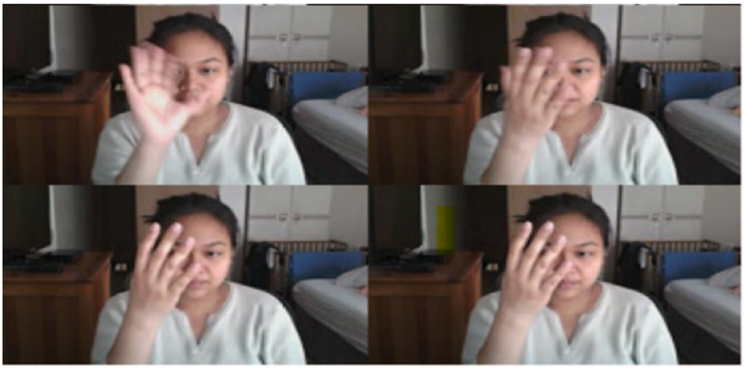

Fig. 3. Examples of videos from the 20BN-jester, a randomly sampled video dataset.

11 hand gestures under 3 different lighting conditions and is released under a Creative Commons Attribution 4.0 license (see Fig. 4). This dataset can be obtained by visiting this link : https://research.ibm.com/interactive/dvsgesture/.

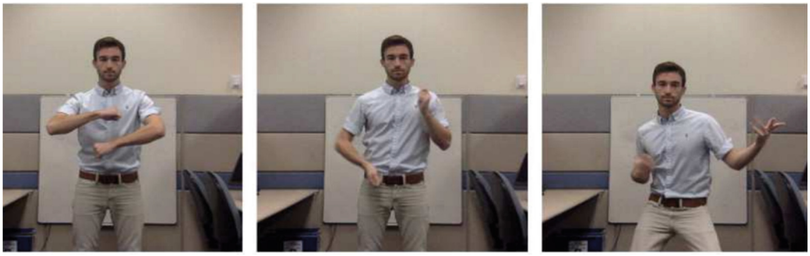

Fig. 4. Dvs -Gesture Dataset video frames of the gestures.

Ego-Gesture Dataset. The Ego-Gesture dataset is a large dataset for recognizing centered hand gestures. The dataset contains 2081 RGB-D videos, 24161 gesture samples, and 2953224 frames from 50 distinct clips. This dataset can be obtained by visiting this link: http://www.nlpr.ia.ac.cn/iva/yfzhang/datasets/egogesture.html. The dataset contains 83 categories of static or dynamic gestures that focus on interacting with devices (see Fig. 5).

ASL Alphabet. The dataset of images of alphabets from American Sign Language is separated into 29 volumes, representing the different categories. 87,000 images with a size of 200x200 pixels. Training data set. 26 volumes are for letters A-Z, and 3 chapters are for SPACE, DELETE, and NOTHING. These last three categories are very useful in applications and real-time classification. Only 29 images are available. The test data is on 29 images only (see Fig. 6A). This dataset can be obtained by visiting this link: https://www.kaggle.com/datasets/grassknoted/asl-alphabet.

Arabic Alphabets Sign Language Dataset (ARASL). A data set that collects Arabic language signals, and the number of images for each category varies from one category

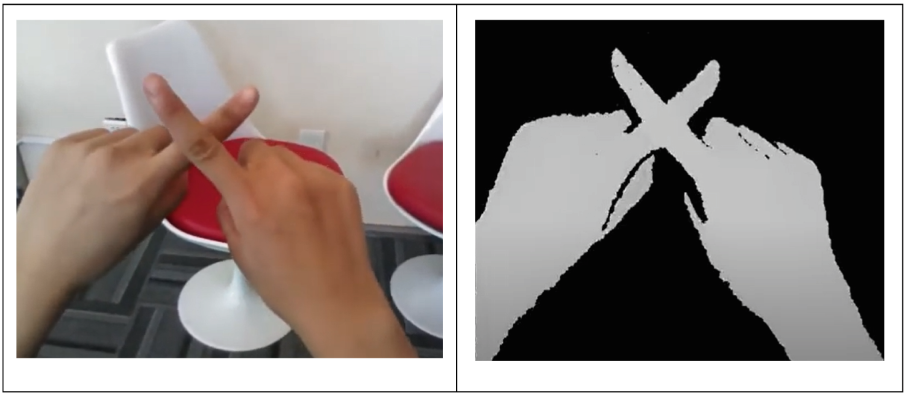

Fig. 5. Sample of Ego-Gesture.

to another. It consists of 54,049 Arabic alphabet images for more than 40 people, in addition to 32 standard Arabic signs and alphabets (see Fig. 6B). This dataset can be obtained by visiting this link: https://data.mendeley.com/datasets/y7pckrw6z2/1 .

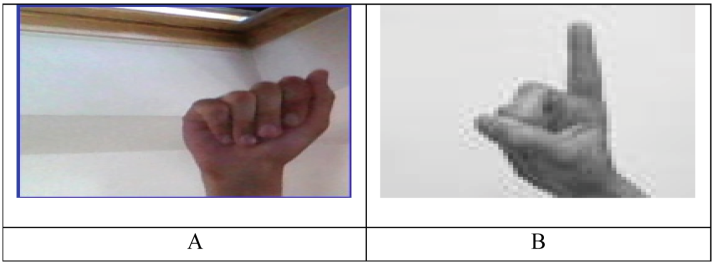

Fig. 6. A- Letter "A" from American Sign Language B- Letter "Baa" from ArASL.

3 Methodology of Research

The topic of hand gesture recognition is vast, and much research has been done in recent years. This study surveys the most recent studies on hand gesture recognition. We will also contrast the various methodologies and applications described in the examined papers. The review indicated that most articles use a single camera webcam or laptop for the data acquisition process.

In recent years, vision-based hand gesture recognition research has shifted to integrating more in-depth information sources like Microsoft Kinect and Leap Motion Controller.

The survey methodology is divided into two parts: the techniques that rely on the principle of artificial intelligence for hand gesture recognition, and the other section is based on the techniques used in artificial neural networks and deep learning for hand gesture recognition, as illustrated (see Fig. 7).

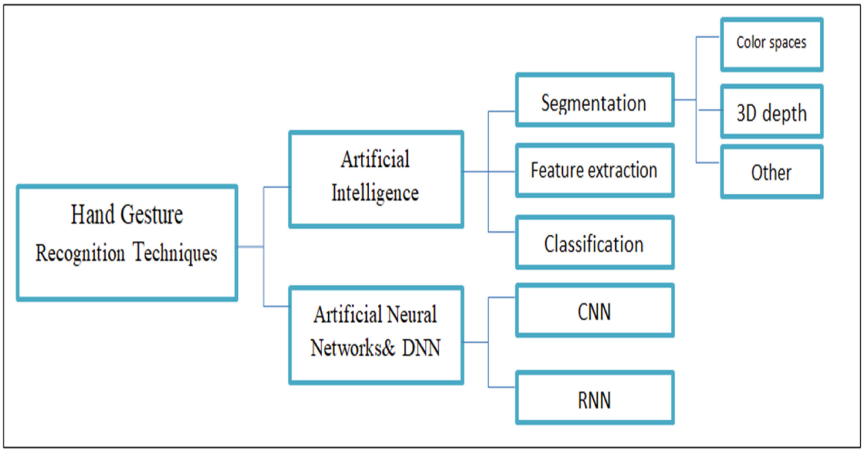

Fig. 7. Survey Methodology

3.1 Hand Gesture Recognition Based on AI Techniques

This section reviews hand gesture segmentation methods and features extracted for diverse applications as well as hand gesture classification algorithms based on AI techniques. The research deals with a serious survey of each stage of hand gesture recognition and focuses on hand gesture recognition systems based on segmentation, extraction of features, and classification, each stage separately (see Table 1).

Segmentation. Image segmentation is one of the most frequently used techniques in image processing. The appropriate method for segmenting images, in fact, depends largely on the quality of the images and the dataset, if it is static or dynamic hand motion. Many methods of segmentation have been dealt with in recognizing hand gestures, starting from discovering edges, clustering, and using threshold or contour methods [20].

Using traditional image processing approaches to recognize the person's hands, such as background reduction based on running average, skin color thresholding, upper body and face cascade classifiers, and finger recognition based on contour detection, the video frames yielded unsatisfactory results.

The research [21] examined three methods for segmenting dynamic hand gestures based on the gesture recognition system. The research suggested both segmentation methods related to skin and movement in addition to contour features. The color space has been changed to Y–Cb–Cr. Morphology operations were also applied after the contour process to obtain the hand area only and to increase the accuracy of segmentation (see Fig. 8).

To encode the color information in an image, a mathematical model can be derived from the color space, which, axiomatically, can be used. There are many different color spaces that can be used for many kinds of applications [22], including computer vision applications, image processing applications, and digital graphics applications.

Table 1. Summary of Review of Techniques for Hand Gesture Recognition.

Ref.	AI techniques			Data set	No. of Gesture	Accuracy	Application Area
	Segmentation	Feature extraction	Classification				
Hernández, I (2018)	Color space Contour& motion	Morphology operator	-	Dynamic Hand Gestures	Unlimited	98%	Sign Language Recognition
Shaik, K (2017)	YUV color	-	-	Hand Gestures	Unlimited	-	HCI
Biswas, A (2018)	fingertips	Hough transform	-	Finger Count	unlimited	-	Sign Language Recognition
Perimal M. et al. (2018)	Y–Cb–Cr	-	-	Finger Count	14	50 to 70	HCI
Huang, H (2019)	Color space	-	SVM	Hand Gestures	9		HCI
ljawaryy, A.et al. (2017)	Y–Cb–Cr	-	-	Count Of Finger	6 Gestures	98%	Deaf, -Mute People
Prakash, J. et al. (2019)	YUV & CAMShift	-	navie Bayes	Hand Gesture	Unlimited	High	Human - Machine Interaction
Zhang, F. et al. (2020)	Landmark point	skeleton pixels	-	Finger Paint	Unlimited	-	Real Time- Hand Tracking
Karbasi, M. et al. (2017)	depth and color space	skeleton pixels	-	Hand Gesture	Unlimited	-	Malaysian Sign Language
Ma, X. et al. (2018)	local neighbor method	convex hull detection	-	Hand	6	96%	Human–Robot Interaction
Kim, M et al. (2016)	Threshold segmentation	-	-	Finger Counting & Hand Gesture	Unlimited	-	Mouse-Movement Controlling

(*continued*)

Table 1. (*continued*)

Ref.	AI techniques			Data set	No. of Gesture	Accuracy	Application Area
	Segmentation	Feature extraction	Classification				
Tekin, B. et al. (2019)	YCbCr, SkinMask, and HSV	-	convolutional neural network	Hand Gesture	Unlimited	95.27%96.33%, and 97.43%,	Sign Language Recognition
Wan, C. et al. (2019)	depth maps	3D hand pose	neural network	Hand Gesture	Unlimited	-	HCI
Ge, L. et al. (2019)	Single RGB image	3D hand shape & pose	-	Finger Paint Dataset	Unlimited	High	HCI
Taylor, J et al. (2017)	mask Kinect body tracker	3D hands shape	-	Hand Gesture	Unlimited	-	Interactions With Virtual And Augmented
Tsoli, A. et al. (2018)	bounding box & hand mask	-	-	Hand Gesture	Unlimited	High	Interaction With Deformable Object & Tracking
Chen, Y. et al. (2019)	feature extraction hierarchical	estimation 3D hand pose	-	Hand Gesture	-	Good	Human–Computer Interaction
Ge, L. et al. (2018)	Depth image	3D hand pose	-	Hand Gesture	-	-	Virtual/Augmented Reality
Wu, X. et al. (2018)	hand joints detection	hand pose	-	Hand Gesture	-	-	(HCI), Virtual And Mixed Reality
Cai, Y. et al. (2018)	-	hand pose	-	Hand Gesture	-	Effectiveness	(HCI), Virtual And Mixed Reality

(*continued*)

Table 1. (*continued*)

Ref.	AI techniques			Data set	No. of Gesture	Accuracy	Application Area
	Segmentation	Feature extraction	Classification				
Ahmed, K. et al (2019)	Machine learning	-	-	Hand Gesture	-	effective method	HCI
Abdul, R. et al. (2017)	Machine learning	-	-	Gesture Controlled	-	Simple and effectiveness	Control MS Windows Via Hand Gesture
Zhou, W (2017)	-	DFT	NN	Static Hand Gesture	-	-	Real-Time Implementation
Liu, Xi. et al. (2017)	Wavelet	wavelet invariant moments	SVM	Hand Gesture	-	Good	HCI
Salunke, T. et al. (2019)	-	HOG	K-NN	Hand Gesture	-	Effectiveness	HCI, Power Point Control
Saha, H. et al. (2018)	-	convex decomposition	-	Sign Finger	-	Good	American Sign Language
Molina, J. et al. (2017)	depth information	motion gesture	Motion patterns	-	26 Gesture	95%	HCI- Virtual Environments
Xi, C. et al. (2018)	YUV	skeleton points	naïve Bayes	Hand Gesture	Unlimited	High	Human And Machine System
Devin, G. et al. (2018)	hand-skeletal joints	-	CNN	-	14 28 Gesture	91.28% 84.35%	HCI

(*continued*)

Table 1. (*continued*)

Ref.	AI techniques			Data set	No. of Gesture	Accuracy	Application Area
	Segmentation	Feature extraction	Classification				
Mujahid, A. et al. (2021)	-	YOLOv3		-	gestures	97.68	HCI
Alnaim, N. et al. (2019)	-	features extraction by CNN	ADCNN	Hand Gestures	7 Hand gestures	training 100% test set 99%”	HCI People-Communicate
Chung, H. et al. (2019)	skin color detection &morphology	deep CNN	-	Hand Gestures	6 Hand gestures	training 99.9% test 95.61%	Home-Control
Bao, P. et al. (2017)	–	deep CNN	-	Hand Gestures	7 Hand gestures	simple backgrounds 97.1% complex background 85.3%	Control A Piece Of Consumer Devices
Li, G. et al. (2019)	skin color	CNN	SVM	Hand Gestures	8 hand Gesture	98.52%	HCI
Wu X,Y (2019)	Gaussian Mixture model	CNN	-	Hand Gestures	7 hand Gesture	95.96%	Hand Gesture Recognition For Human

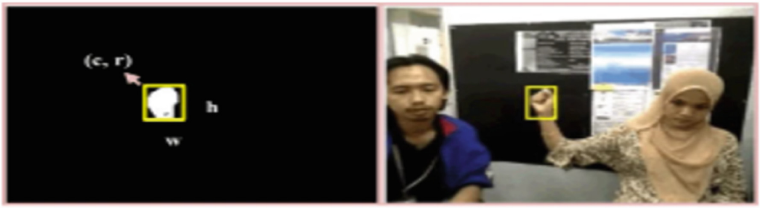

Fig. 8. Segmentation result in [21].

Fig. 9. Skin Color Detection. (A) Use The Color Space YUV Threshold. (B) Detect and Track the Hand Using the Resulting Binary Image [22].

(See Fig. 9) is a demonstration of the YUV color space being used to identify different skin tones. [23] They used the Hough transform to extract features for the fingertips.

A new technique that describes the number of hand gestures ranging from 0 to 5 that people indicate that a computer understands has been proposed by [24]. This technique is implemented by reading a frame as an image and then extracting it only manually using the YCbCr color space filter. Then it is converted to a black and white image. The experiment was conducted using 180 random hand gesture frames taken from random people (see Fig. 10).

It was found that under-regulated lighting circumstances, high-resolution cameras, and a short distance away, gestures were examined with three criteria that directly affect the identification rate, namely noise, light levels, and hand size. The method conducts finger counting and gesture recognition based on the maximum distance between the observed fingers, with an average performance of 50 to 70% of successful detection for 14 movements based on color space-based Y–Cb–Cr segmentation. The Y–Cb–Cr color space was shown to be helpful. Lighting effects that impede detection are particularly difficult to remove.

Skin color detection using RGB channels in [25]. Because the blend of an image's color channel and intensity information has uneven features, it is not recommended for skin segmentation. Because the threshold value is an average of three-channel values, adjusted RGB color information is isolated from the brightness in a straightforward manner (r, g, and b). However, it cannot be depended on for segmentation or detection under changing illumination conditions. Converting the RGB color space to another color space in order to identify the region of interest in the hand through other hues such as red, pink, orange, and brown is an important procedure. The color may be determined. The hue of the sample may be predicted ahead of time. If it is comparable, we might

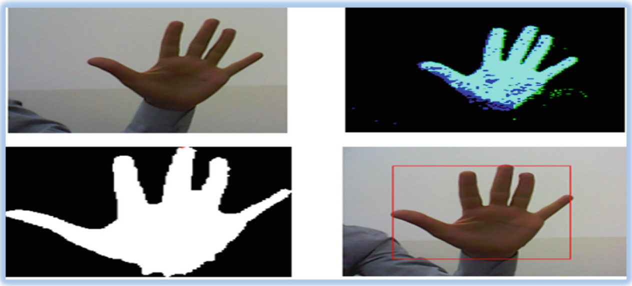

Fig. 10. Hand Gesture Detection and Foreground Segmentation Stages [24].

refer to it as the skin or the region of interest. The skin color approach has a number of obstacles, including lighting fluctuation, backdrop concerns, and other sorts of noise.

[26] found that applying the Y–Cb–Cr color scheme helps to eliminate lighting effects, but that intense light during capture affects accuracy. With the help of the CAM Shift algorithm, YUV color space was used in [27]. The background color was distinguished from skin color, and then a naive Bayes classifier was used to aid in gesture recognition.

The hand segmentation algorithm used in the research [28] was performed using a Kinect camera depth sensor, followed by a calculation of fingertip locations using 3D communication, Euclidean distance, and geodetic distance on manual skeletal pixels to provide higher accuracy.

In studies [29], a hybrid technique of hand segmentation utilizing depth and color data from the Kinect sensor has been suggested. In addition to relying on skeletal data, in the suggested technique, the image threshold was applied to the depth frame as a pixel segmentation technique to extract the hand from the color frame.

Several academics have proposed the use of a depth camera that offers three-dimensional geometric data about the item [30, 31]. This is due to the fact that the three-dimensional data from the depth camera directly reflects the depth field and is unaffected by lighting, shadows, and color, unlike a color image, which only comprises projection. However, the cost and accessibility of depth cameras will limit their application. In [32], they utilized Y-Cb-Cr, SkinMask, and HSV segmentation techniques (hue, saturation, and value). The SkinMask procedure is then subjected to binarization and color segmentation in order to identify pixels that match the color of the hand. Threshold masking determines the dominating characteristics in the HSV process (see Fig. 11).

Studies by [33–35] proposed a model that used 3D hands and objects using a single RGB image to understand interactions between them. [36–40] presented a method for tracing a complex deformable object by interacting with the hand based on the 3D position of the hand in single-depth images. Artificial Neural Networks (ANN), Fuzzy Clustering Algorithm, Graph-based Feature, Hidden Markov Model (HMM), Condensation Algorithm, and Finite State Machine (FSM) are all mentioned methods. In [41] as an effective method to recognize hand gestures.

Feature Extraction. The accuracy of the quality of any algorithm depends on the quality of extracted features, because these features, or rather important information, give a

Fig. 11. Segmentation in [32]

complete impression of the classification or recognition process. Therefore, accurate information extraction from images must aid in the process of gesture recognition.

The Fourier descriptor, and fingertips, contour hand, motion gesture, or centroid, capture the basic structure. It can be considered a common method of feature extraction. Both contour hand and complex Alhzat are feature extractions used in [42]. In [43], employing Discrete Fourier-Transform (DFT) techniques on histograms, a method for extracting features was implemented (vertical and horizontal).

In [44], the wavelet feature is determined by enforcing the wavelet invariant moments of the hand region, and the distance feature is extracted by calculating the distance from the fingers to the hand centroid. Both of these features are determined by calculating the wavelet invariant moments of the hand region. After that, a feature vector is created, and it is made up of wavelet invariant moments and a distance feature. Last but not least, a support vector machine classifier that is based on the feature vectors.

The method that was proposed in [45] first processed the images after extracting a histogram of gradient features from the input data that was provided by a portable webcam. The portable webcam provides the proposed system with the input data, which comprises four different hand motions to be interpreted. After the image that was acquired has been processed using the data that was entered, the graph of the vector gradient features will be retrieved from the obtained image. Following this, the processed image is looked up in a database that contains images of gestures. The K-Nearest Neighbor method is used to perform the comparison and recognition of the image. The slide show is then controlled by the image that was initially detected.

American Sign Language was used in the study. [46] Images are collected from the webcam and preprocessed. The figure is divided into two parts by means of polygon approximation and convex decomposition approximation. Features are extracted by recording the unique features between the different convex parts of the hand. The resulting singularity was used as the extracted feature vector. This training includes features that are almost unique to various hand gestures.

This article [47] describes current machine learning approaches for recognizing hand gestures using detailed data from time-off light sensors. Experiments have shown that convolutional neural networks and long-term memory provide the most reliable results for implicitly extracted features. It is a method that uses image color statistics and also uses a “regional contrast” (RC)-based extraction algorithm for prominent objects.

[48] used a different technique to extract hand features, namely a histogram of oriented gradients (HOG) of oriented gradients (HOG) was used for image processing to extract characteristics of the hand. The Accurate End Point Identification method was implemented and applied to gesture images which were captured in varying backgrounds to detect edge points and branch points, and it was also applied to blurred images containing multiple objects.

[49] A three-dimensional convolutional neural network model for learning region-based spatiotemporal aspects of hand movements has been proposed. This model takes as its input a sequence of RGB frames that were taken by a straightforward camera. In order to extract the features from the full video sample, a 3DCNN was used, and a SoftMax layer was utilized in order to perform the classification. It acquired recognition rates of 84.38%, 34.9%, and 70% on the three datasets, respectively, by using three gesture datasets derived from color videos: KSU-SSL, ARSL, and ASL classes.

Higher-order local autocorrelation (HLAC) features are used to extract features in the novel method of feature extraction that is described in [50]. This method is a recent development in the field of feature extraction. Identifying the texture of the image of hands in relation to prospective location and the pixel product picked in white requires the extraction of features from grayscale images using a variety of masks. Then, based on the Mutual Information Quotient, the characteristics that hold the most valuable information are chosen for selection (MIQ). In order to categorize the various ways in which people use their hands, a multiple linear differentiation analysis (LDA) classifier was utilized. Experiments conducted on the dataset provided by NUS reveal that HLAC is superior to other feature extraction approaches in terms of its ability to recognize hand motions.

In each of the following works [51] and [52], several methods were used to extract specific features and apply them to gesture images. The first research focuses on the identification of accurate endpoints, which are applied to gesture images to detect edge points and branch points, as well as in blurry images that contain multiple objects. The second study presents a double-channel convolutional neural network (DC-CNN) in which the original image is preprocessed using canny detection to detect the edge of the hand before being given to the network. To categorize output results, each two-channel CNN has its own weight and SoftMax classifier.

In order to obtain the hand edge images, first the original gesture images must be preprocessed, and then edge detection must be done on those images. Second, the hand gesture images and the hand edge images are each chosen to serve as one of CNN's input channels in their respective orders. The number of convolutional layers and the other parameters are identical across all channels. However, each channel has its own unique weight. The final step is to execute feature fusion at the complete connection layer, and the output result is then classed using the SoftMax classifier. There are ten different gestures stored in the database.

In [53], depending on depth information, feature extraction was motion gesture and compared the motion patterns for 26 gestures. In [54], a Kinect camera depth sensor was used to capture the image. Euclidean distance and geodesic distance fingertip skeleton pixels were extracted. By using a depth camera, the feature extraction based on hand-skeletal joints' positions data set is dynamic hand gesture [55]. In [56], based on Kinect camera features, gestures were extracted by computing skeleton point clouds with a YOLOv3 algorithm, while [33, 34, 37, 38, 40] feature extraction depended on using 3D hand pose estimation.

Classification. There are many classification algorithms that can be used to identify different gestures, whether the gesture is dynamic, sign language, etc. The extracted features are sent to one of the classification methods, and the data is divided for testing and training the classification algorithm, through our survey of a number of researchers. The result was that the most used methods for classification are both Artificial Neural Networks (ANN), K-Nearest Neighbor (KNN), Naive Bayes (NB), Support Vector Machine (SVM), etc.

The study [57] provided an effective, fast, and easy method for dynamic hand gesture recognition based on two-line features extracted from two-line real-time video. The feature selection was used to represent the hand shape of the Kurdish Sign Language's dynamic word recognition. Features extracted in real-time from preprocessed manual objects were represented by optimized values for the captured binary frames. Finally, we used an artificial neural network classifier to recognize hand gestures.

[58] Hand gesture recognition algorithms for the elderly are offered. Using contour and rule-based algorithms, respectively, detection, feature extraction, and classification are the three main components of the procedure. The method relies on the vision-based recognition of six fixed hand gestures.

In [32], they used the segmentation methods of YCbCr, SkinMask, and HSV, then binarization. SoftMax classification is used to classify hand gestures where features are extracted through convolutional neural networks (CNN). While in [49], a SoftMax layer was used for classification. Using three gesture datasets from color videos: KSU-SSL, ARSL, and ASL classes, it obtained recognition rates of 84.38%, 34.9%, and 70% on the three datasets, respectively. Also, [52] used SoftMax for classification.

They were able to identify hand gestures in their research [59–62] by utilizing the Nearest Neighbor (KNN) classifier. The work that was presented in reference number [63] identified detections by employing a hybrid of the k-nearest neighbor (KNN) and decision tree classification methods. The classification of [64–68] gestures was also accomplished with the assistance of a Support Vector Machine (SVM). The gestures of [69] were organized into categories using a decision tree.

They employed a tweaked version of the deep forest algorithm in order to identify the pattern in [70]. The classification process in [71] was carried out using a random regression forest technique. The hand gesture is modeled and decomposed using the Gaussian mixture model and Hidden Markov Model, and the GMM is employed as a substrate of the HMM to decode the sEMG function of the gesture. This can be found in [72]. In addition, a hand gesture detection system was developed and implemented by introducing Bhattacharyya divergence into the Bayesian Sensing Hidden Markov Model (BSHMM). This research was published in [73].

Other. Far from using the segmentation algorithm, [74] proposed a real-time dynamic hand gesture detection system based on TOF. This system was designed to work in real-time. The act of identifying movement patterns based on hand motions captured as input depth images is referred to as "pattern recognition. The system works well overall, except that the depth of field of the TOF camera isn't good enough.

A new method is presented in [75] for RGB video recording of sign language based on extracted structural features, in addition to taking into account the difficulties in extracting accurate structural data due to obstruction. Using skeletal and depth data for a skeleton-based approach [76], a dynamic and recognizable hand gesture was introduced (see Fig. 12).

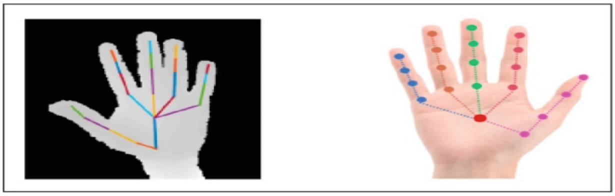

Fig. 12. Skeleton Dataset to Representation Hand Skeleton Model [76].

Using Kinect sensor depth metadata, a new dynamic hand gesture recognition technique was described in [77]. For classification and recognition, an orientation feature retrieved for segmentation (SVM) algorithm and HMM were used.

A three-part gesture recognition robot system was introduced [78]: an intelligent robot system, gesture recognition, and an Android app. Three independent discriminators contained in the gesture recognition system were verified to control gesture recognition based on the use of an algorithm based on feature extraction and template matching. The average rate of gesture recognition was 94.6%.

Principal Component Analysis (PCA) and classifiers such as the Support Vector Machine (SVM) and k-near-neighbor (k-NN) were used in addition to the recognition experiments. Reducing this similarity between frames is an important step in pattern recognition. The method was applied to the Irish Sign Language data set [79].

This article [80] focuses on a dynamic 3D gesture recognition system that uses hand location information. It uses the natural structure of the hand topology (later called hand skeletal data) to extract effective manual kinematic descriptors from a set of gestures. The descriptors are then encoded into statistical and temporal representations using the Fisher kernel and a multi-level time hierarchy, respectively. The proposed approach was applied to a three-handed gesture dataset, each containing 10, 14, and 25 gestures. Also, the proposed evaluation is from the perspective of hand gesture recognition and low-latency gesture recognition (see Fig. 13).

This work summarizes the technologies of the above works by surveying a number of artificial intelligence techniques for recognizing hand gestures (see Fig. 14).

3.2 ANN and DNN Techniques for Hand Gesture Recognition

Even if a related dataset is used to pre-train the network and take advantage of transfer learning, one of the primary challenges associated with deep learning-based image

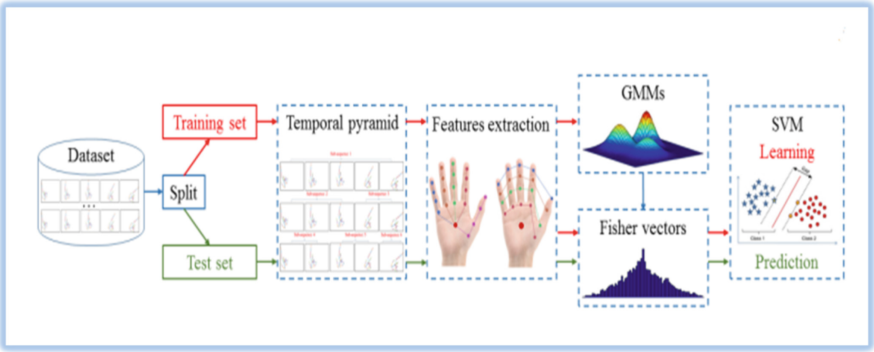

Fig. 13. Overview of The Approach in [80].

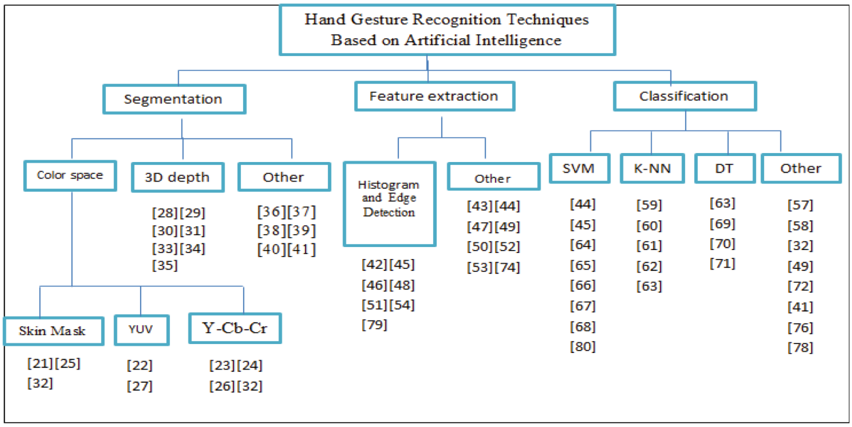

Fig. 14. Taxonomy of Artificial Intelligence Techniques based on Hand Gesture Recognition

recognition is the requirement for a vast amount of labeled data. In addition, there are other issues, such as design optimization, which can be difficult and time-consuming [81]. These challenges are one of the main problems with deep learning-based image recognition. (See Fig. 15) for the research and techniques that have been used recently to recognize hand gestures.

Method Based on CNN. In this paper [82], a standard dataset called IPN Hand is provided. This dataset contains more than 4,000 gesture samples and 800,000 RGB frames from 50 distinct targets for 13 different static and dynamic gestures focusing on interaction with non-touch screens based on 3D-CNN technologies.

Graph CNNs are employed in the study of [83] to create a 3D network of the hand surface. Hand segmentation and 3D hand position estimation, including skeletal constraints, can be learned using a CNN-based approach developed by [84]. Deep learning models using parallel convolutional neural networks (CNNs) to process the positions of hand skeleton joints have been introduced in [85] for the purpose of 3D hand gesture detection.

For classification, a network containing an RNN and additional accelerometer data was employed (CNN), and an RNN was used for the classification. [88] Hand motions were classified using the Deep Convolutional Neural Network. The Deep Convolutional Neural Network (ADCNN) feature from this paper was taken and used for hand categorization in this article. The experiment uses 100% training data, 99% test data, and a total run time of 15,598 s [90].

Another suggested method is to track hands using webcams. The background is then removed using the skin color (Y-Cb-Cr color space) and morphology approach. A kernel correlation filter (KCF) is also employed to monitor the ROI. The final images are sent to the Deep Convolutional Neural Network (CNN). It compared the performances of two modified models, AlexNet and VGGNet, using the CNN model. In [91], the recognition rates for training and test data are 99.90% and 95.61%, respective.

A new method based on the Deep CNN feeds resized images directly to the network, ignoring segmentation and recognition levels and directly classifying hand gestures. The system operates in real-time, with a result of 97.1% for simple backgrounds and 85.3% for complex backgrounds [92].

The depth image generated by the Kinect sensor was used to segment the color image and then combined with a convolutional neural network that applied an error backpropagation algorithm to change the thresholds and weights of the neural network. Use skin-color modeling. An SVM classification algorithm has been added to the network to improve the results [93].

[94] Developed A 2-Channel Convolutional Neural Network (DCCNN) in another research paper, where the original images are preprocessed to detect the edges of the hand before being transmitted to the network. Each of the two-channel CNNs has its own weight, and the output is classified using the SoftMax classifier. The detection rate of the suggested system is 98.02.

RNN-Based Method. The author of [86] has developed two channels for dynamic hand gestures. The detection method uses a low-complexity recurrent neural network (RNN) algorithm for portable devices. It is initially based on video signals and uses convolutional neural networks. In contrast, [87] uses a bidirectional recurrent neural network (RNN) with skeletal sequence to enhance motion, and an RNN function was used.

In this paper [89], an improved recurrent neural network is proposed for skeletal-based dynamic hand gesture recognition. After finger movement and general movement features are extracted, they are fed into a bidirectional recurrent neural network (RNN) along with the extracted finger movement features to describe finger movements. This method is effective and superior to other art initiation methods.

The hand gesture recognition system successfully identifies 10 gestures in [96]. The entire system includes a signal processing part that produces a Range-Doppler Map

(RDM) sequence without clutter, and a machine learning part that is an LSTM (long-term memory) encoder for learning the temporal characteristics of the RDM sequence.

Hybrid Method. This study [95] proposes combining the capabilities of two deep learning technologies, a convolutional neural network (CNN) and a recurrent neural network (RNN), to automatically recognize hand gestures using depth and structure data. The purpose of this study is to use depth data and apply CNNs to extract important spatial information from images. Tandem CNN + RNN can recognize different gestures. It combines both skeletal and depth information to extract temporal and spatial information.

A proposed deep learning-based method for temporary 3D shape recognition issues [97] is primarily based on a combination of a convolutional neural network (CNN) and a long-term memory (LSTM) recurrent network. Introducing a two-level education method that focuses on CNN education at first and then changes the whole method (CNN + LSTM) at the second level.

This paper [98] affords a powerful deep structure for non-stop gesture recognition. First, continuous gesture sequences are broken up into remoted gesture times using the proposed temporally dilated Res3D community, which is mostly based on the 3-D convolutional neural community (3DCNN), the convolutional long-short-term-memory network community (ConvLSTM), and the 2-D convolutional neural community (2DCNN) for remoted gesture recognition.

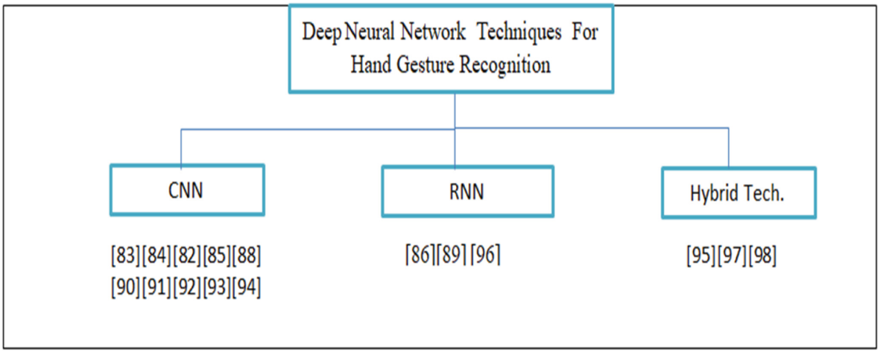

Fig. 15. Taxonomy of Deep Neural Network Techniques for Hand Gesture Recognition

4 Conclusion

There is no doubt that understanding various hand gestures, whether they are sign language or other expressions, is critical. Researchers have made great strides in hand gesture recognition because of its widespread practical applicability. In light of this, it is crucial to highlight the most cutting-edge technologies in use by conducting a comprehensive literature review on the subject of vision-based hand gesture recognition, with a particular focus on those based on the principles of artificial intelligence and deep learning techniques. The ability to use these technologies in a wide range of critical applications makes computer communication more natural. There are several factors that affect the interpretation of gestures, including the background, the source of light, and the skeletal structure of hand movement. The majority of studies have found that edge detection, depending on geometric features for fingers, is particularly useful for recognizing gestures when the fingers of the hand are extended but that recognition is more challenging when the hand is closed. Recently, deep learning has seen widespread application in a lot of research for hand gesture recognition with high accuracy. Through the comprehensive research survey, the associative memory network has not been used in any of the surveyed research. It is necessary to work on techniques based on neural networks or deep learning, which give better accuracy of results than other traditional methods.

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